Data Provided: None

DEPARTMENT OF COMPUTER SCIENCE

Computer Vision - Practice Test

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1. These questions are about Filtering, Clustering, Segmentation and Edge Detection.

   a) What is true about quantization? (single-answer) [20%]

   (A) Affects image resolution
   (B) Refers to the possible values used to represent a sample
   (C) Is related to how many readings (in space) are made from a scene
   (D) Is related to how many sensing elements the camera sensor has

   b) What is true about cross-correlation? (multiple-answer) [20%]

   (A) Is a linear operation
   (B) Using a vertically and horizontally flipped kernel, corresponds to convolving using the original filter
   (C) Is useful to see how similar two signals (i.e. images) are
   (D) Is used in the median filter

   c) Which point operator can be used for a fast increase of the low (i.e. dark) intensities of an image, and how? [20%]

   d) When using the gradient information for edge detection, what does the gradient magnitude tell us? What does the gradient direction tell us? How does the Canny operator use the gradient information? [40%]

2. These questions are about Feature Detection, Description, Transformation, and Camera Model.

a) What are the characteristics of a good feature in an image? (multiple-answer) [20%]

(A) Detectable under geometric and intensity changes.
(B) Possesses distinctive properties.
(C) Looks pretty.
(D) Uses a large quantity of information around it.

b) Select all the parameters of the matrix of intrinsic: (multiple-answer) [20%]

(A) Focal distance.
(B) Skew.
(C) Offset from the center (of the projection plane).
(D) Rotation.

c) The planar facade of a building is captured in an image. Assume a scene point (X, Y) on the building projects to image pixel coordinates (u, v) by the below projective transformation:

$$\begin{bmatrix} wu \\ wv \\ w \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ p_{31} & p_{32} & p_{33} \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix}$$

[20%]

(i) How many degrees of freedom does this transformation have, and explain why?

(ii) How many point correspondences are required to determine this transform, and explain why?

(iii) Would having more correspondences than your answer to (ii) be helpful? If yes, explain how they could be used. If not, explain why not.

d) The SIFT descriptor is a popular method for describing selected feature points. Assuming feature points have been detected using the SIFT feature detector, (A) briefly **describe** the main steps of creating the SIFT feature descriptor at a given feature point, and (B) **name** three scenes or image changes that the SIFT descriptor is invariant to. [40%]

3. Sheff & Ships, a pioneering space exploration startup, has successfully deployed multiple imaging satellites into Mars' orbit. They've brought you on board as a deep learning computer vision consultant to develop an advanced mapping system for the Martian surface. The primary goal is to meticulously identify all materials, rocks, and substances present on the surface to evaluate the potential for of colonies on Mars. In summary, for every pixel in an image, the system must classify it into predefined categories established by the company's geologists, such as quartz, magnetite, water, and silicates. The desired output is an image of the Martian surface with the same dimensions of the input with polygons indicating areas with every mineral. Sheff & Ships possesses powerful supercomputers to host the new software and a robust team for data annotation. Our main directive is to achieve maximum accuracy in mineral classification.

   a) If you are required to design a deep learning-based solution for this problem, which elements below would you choose as essential (select up to five)? [30%]

      (A) Convolutional Layers
      (B) Batch-Normalization Layers
      (C) Average Pooling
      (D) Max Pooling
      (E) Fully Connected Layers
      (F) Cross-Entropy Loss
      (G) Transpose Convolution Layers
      (H) Dilated Convolution Layers
      (I) Residual Connections
      (J) Sigmoid Activation Functions

   b) Which of the deep learning-based existing architectures below would be most suitable for this problem? **Justify** briefly. [30%]

      (A) YOLO (B) AlexNet (C) LeNet (D) U-Net (E) R-CNN (F) DenseNet (G) ResNet (H) Efficient-Net

   c) Briefly **describe** your solution to the mentioned classification problem using and properly connecting elements/concepts you have learned in the Computer Vision Module. Remember to mention which kind of data augmentation strategies you would apply to your data, if needed. [40%]

4. Regarding deep learning-based object detection:

   a) One of the simplest ways to perform detection involves the classification of sliding windows. Why does this strategy become unfeasible if it is not implemented efficiently? [30%]

   b) How do 1x1 convolutions work? How have they contributed to the success of object detection architectures like Yolo? [40%]

   c) Object detection methods take advantage of the Non-Max Suppression algorithm and Intersection Over Union (IoU) metric to select candidate windows and eliminate windows that are less likely to contain objects of interest. Suppose we have detected bounding boxes with their corresponding confidence scores and IoU values as follows:

      Box 1: Confidence Score = 0.85, IoU with Box 2 = 0.6
      Box 2: Confidence Score = 0.75, IoU with Box 1 = 0.6, IoU with Box 4 = 0.7
      Box 3: Confidence Score = 0.90, IoU with Box 4 = 0.4
      Box 4: Confidence Score = 0.80, IoU with Box 2 = 0.7, IoU with Box 3 = 0.4, IoU with Box 5 = 0.8
      Box 5: Confidence Score = 0.95, IoU with Box 4 = 0.8

      If the IoU threshold for NMS is set to 0.6, how many boxes will remain after applying NMS? [30%]

END OF QUESTION PAPER